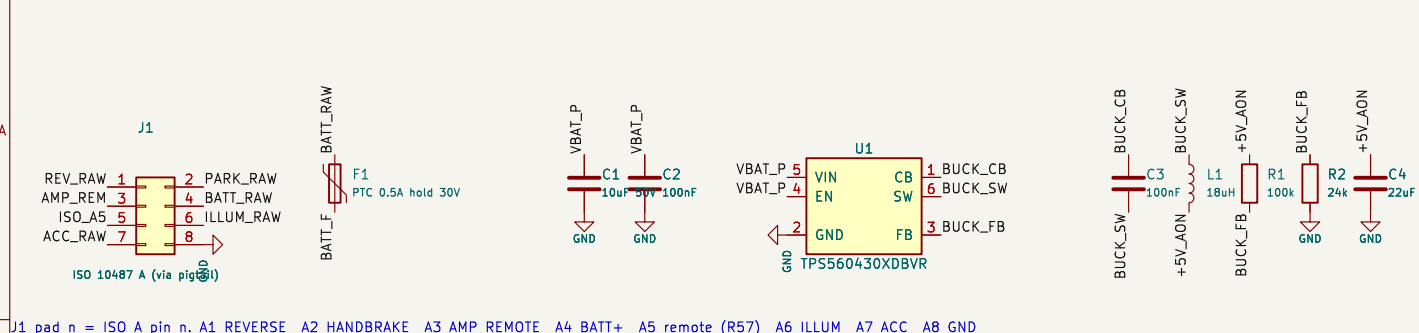
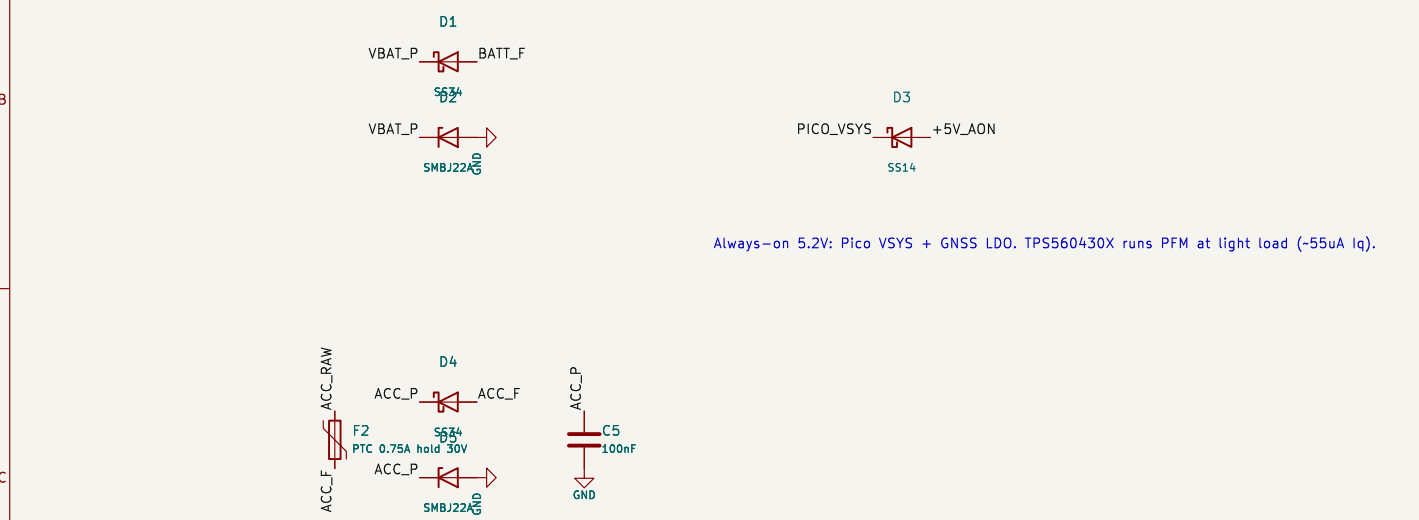


1. ISO 10487 CONNECTOR A (power) + ALWAYS-ON 5V FOR PICO AND GNSS

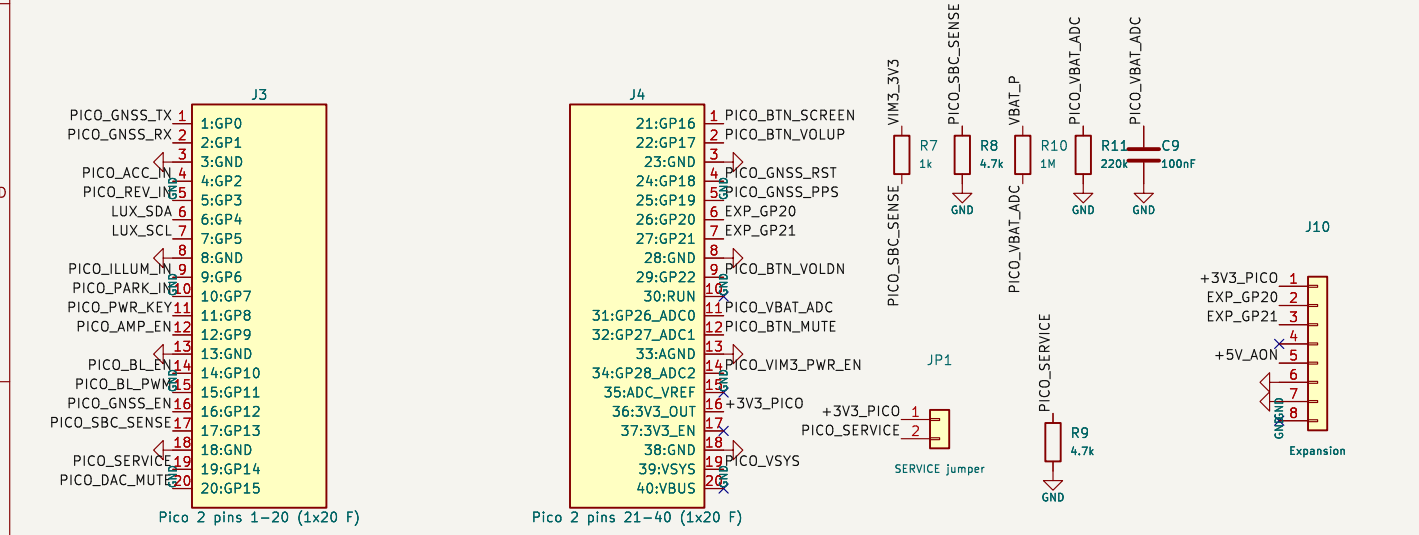


J1 pad n = ISO A pin n. A1 REVERSE A2 HANDBRAKE A3 AMP REMOTE A4 BATT+ A5 remote (RST) A6 ILLUM A7 ACC A8 GND



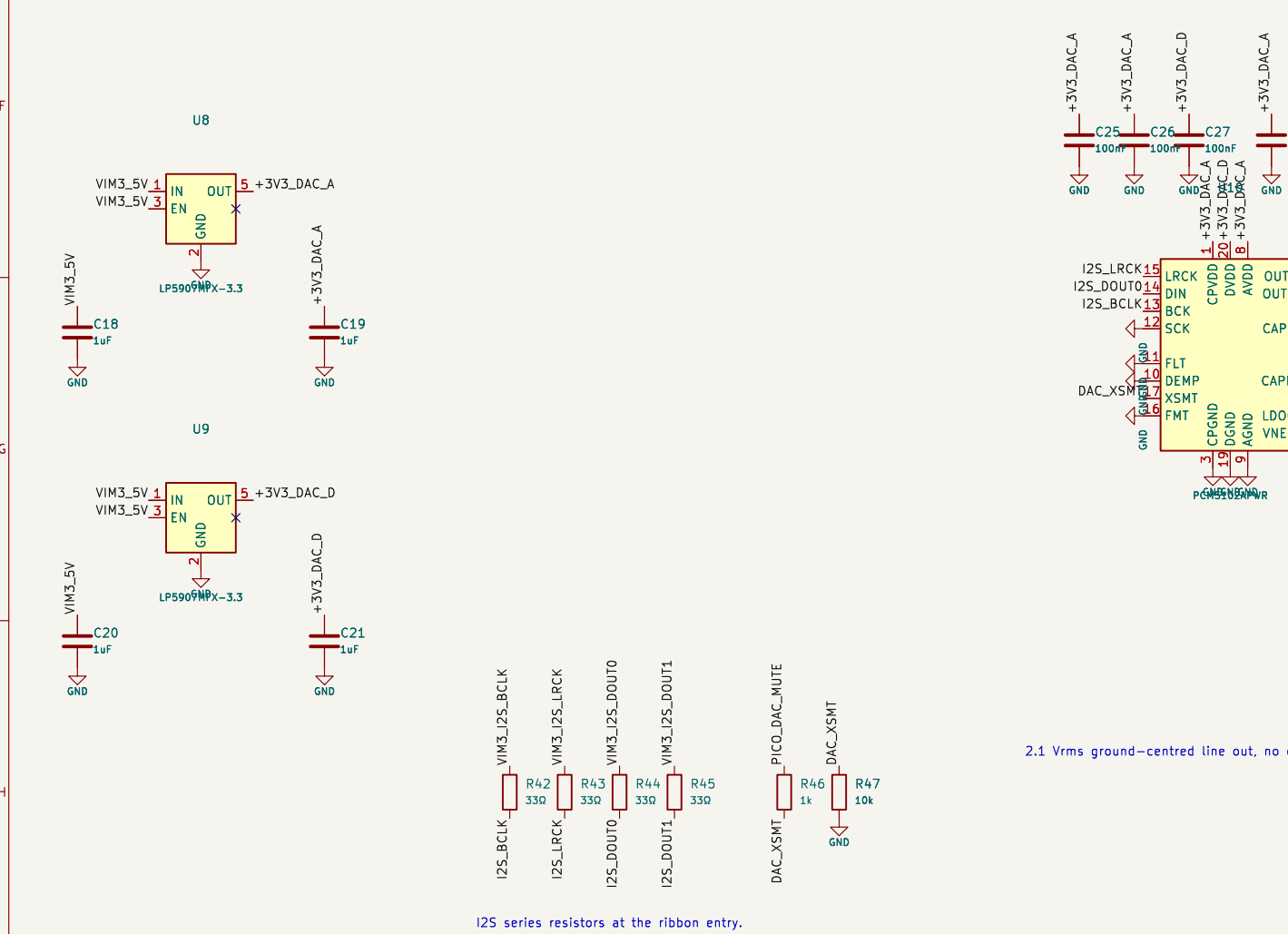
Always-on 5.2V: Pico VSY5 + GNSS LDO. TP555043DX runs PPM at light load (~55uA req).

4. PICO 2 (on two 1x20 sockets) + USB to VIM3



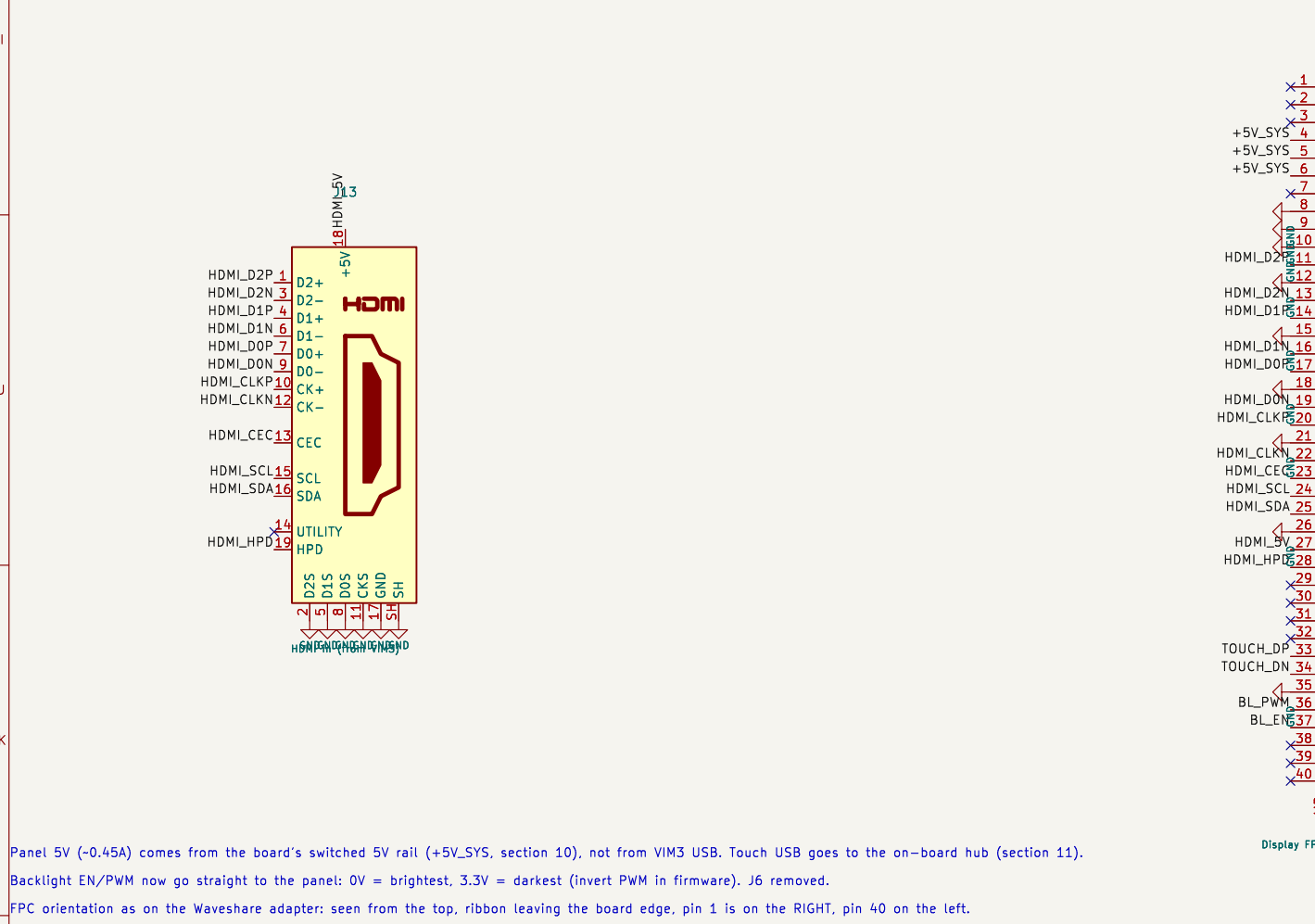
USB: Pico micro-USB cable to a VIM3 USB-A port (HID vehicle data = CDC GNSS bridge).
GP01/ GNSS UART0, GP2 ACC, GP3 REV, GP4/5 I2C I/O, GP6 ILLUM, GP7 PARK, GP8 PWR key, GP9 AMP.
GP10 BL_EN, GP11 BL_PWM, GP12 GNSS_EN, GP13 SRC sense, GP14 service, GP15 DAC mute, GP18 GNSS reset.
GP19 GNSS PPS, GP20/21 UART1 (K-line, reserved), GP16/17/22/27 display buttons, GP28 VIM3 supply enable, GP26 battery voltage.

7. AUDIO: VIM3 TDM-B I2S -> 2x PCM5102A (front + rear) -> line out on ISO C1



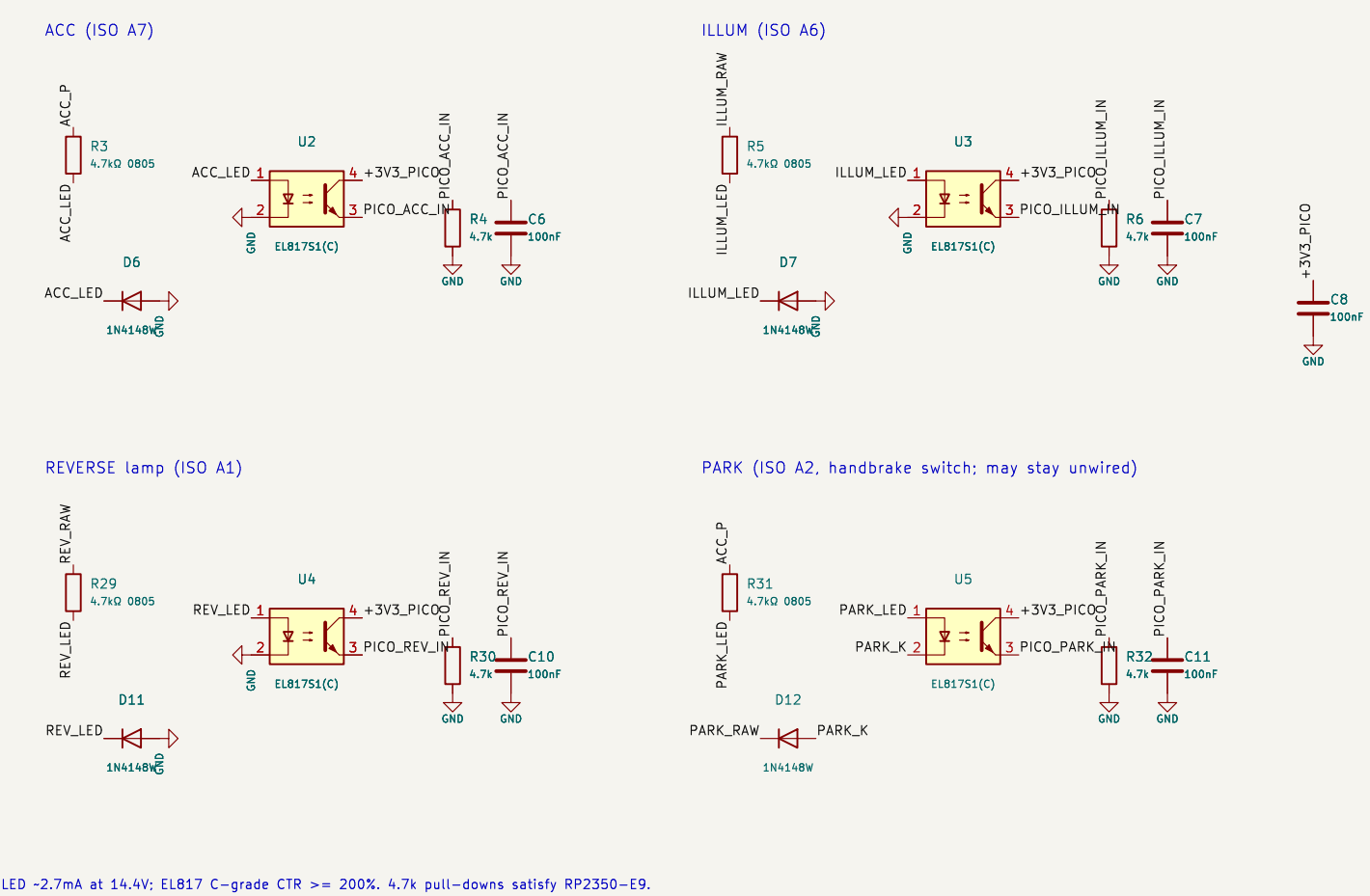
21 Vrms ground-centred line out, no coupling caps. 24-bit in 32-bit I2S slots, 48/96/192 kHz.

9. DISPLAY: Waveshare 7in 70H-1024600 adapter built in (HDMI in from VIM3 -> 40-pin FPC to the panel)



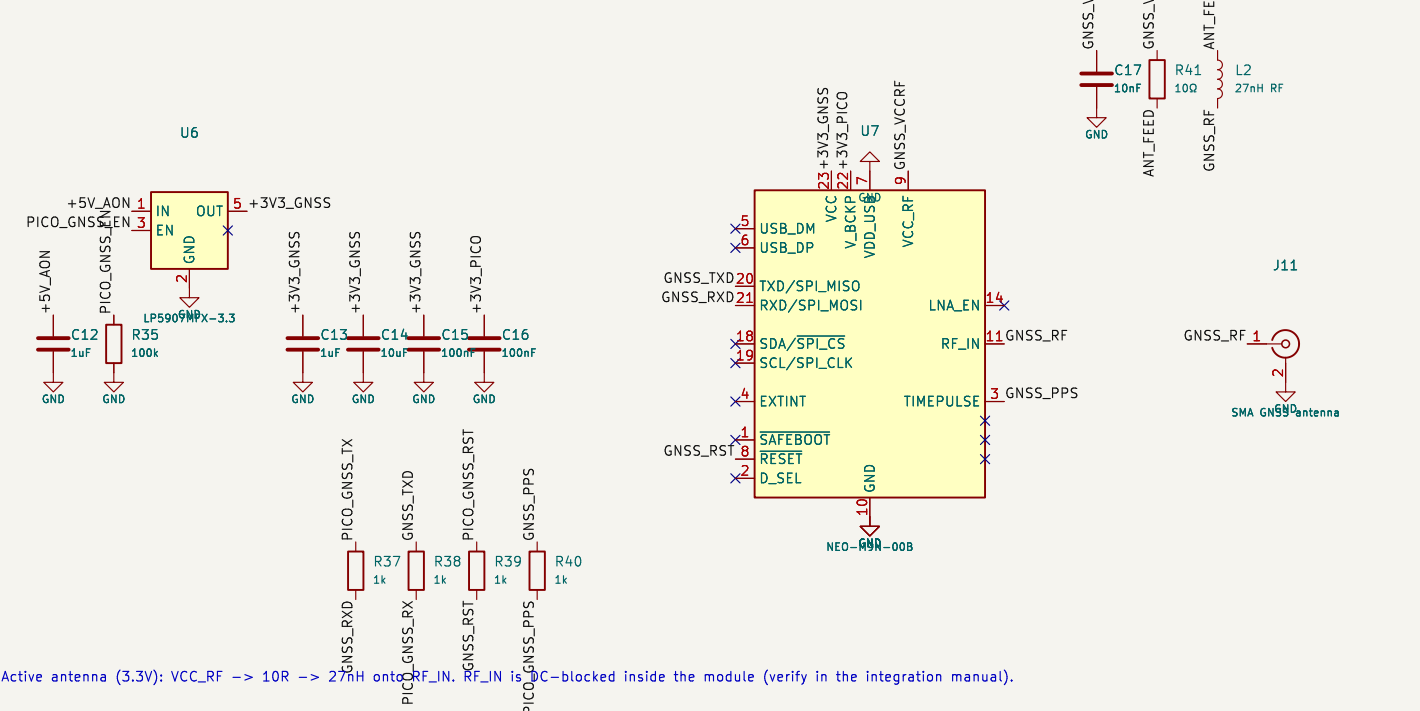
Panel 5V (-0.45A) comes from the board's switched 5V rail (+5V_SYS, section 10), not from VIM3 USB. Touch USB goes to the on-board hub (section 11).
Backlight EN/PWM now go straight to the panel: DV = brightest, 3.3V = darkest (Invert PWM in firmware). J6 removed.
PC orientation as on the Waveshare adapter: seen from the top, ribbon leaving the board edge, pin 1 is on the RIGHT, pin 40 on the left.

2. VEHICLE INPUTS (opto-isolated). ACC, ILLUM, REVERSE: +12V = on. PARK: switch to ground = on.



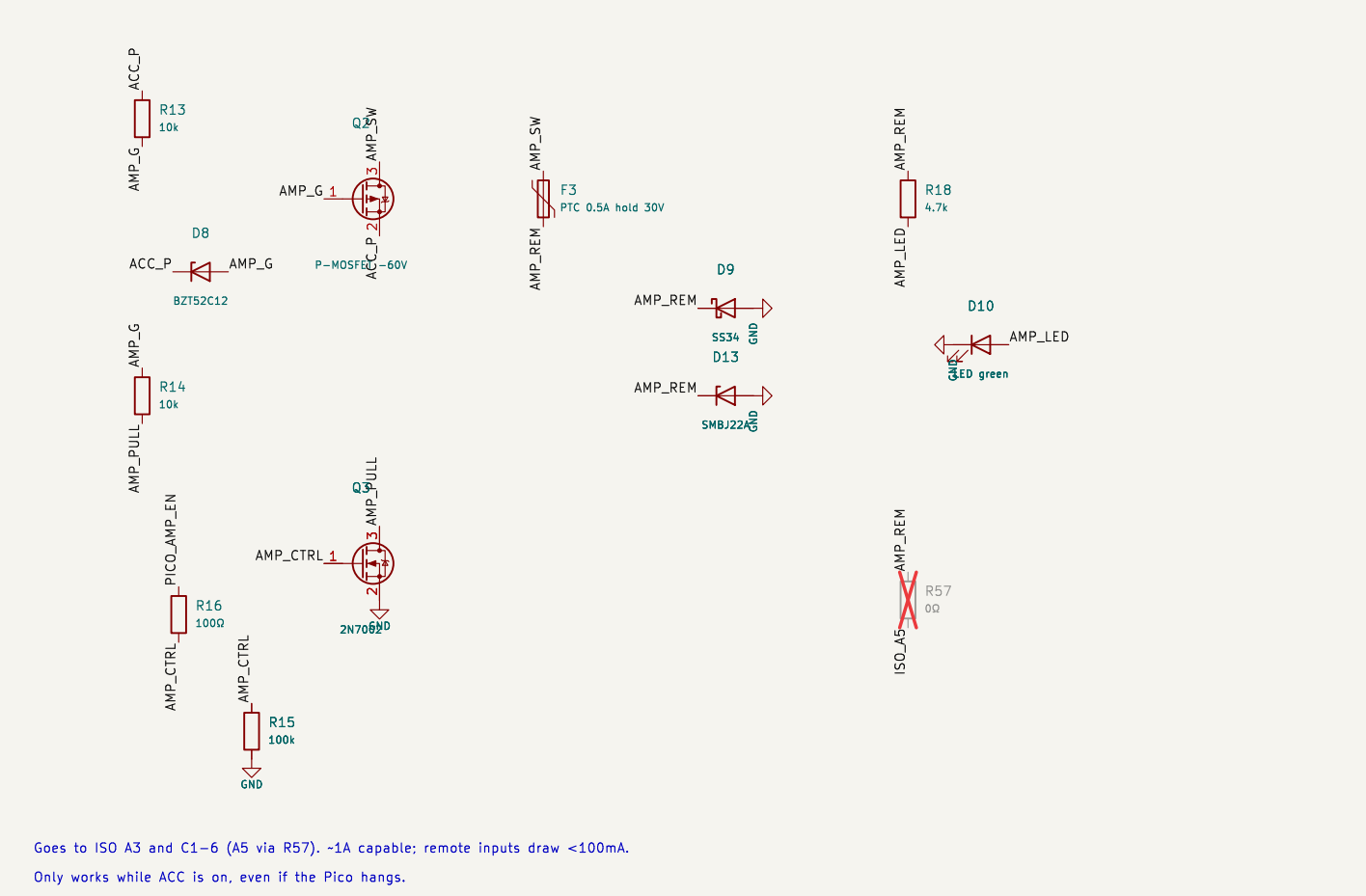
LED ~2.7mA at 14.4V; EL1817 C-grade CTR >= 200%, 4.7u pull-downs satisfy RP2350-ES.

5. GNSS: u-blox NEO-M9N, active antenna on SMA



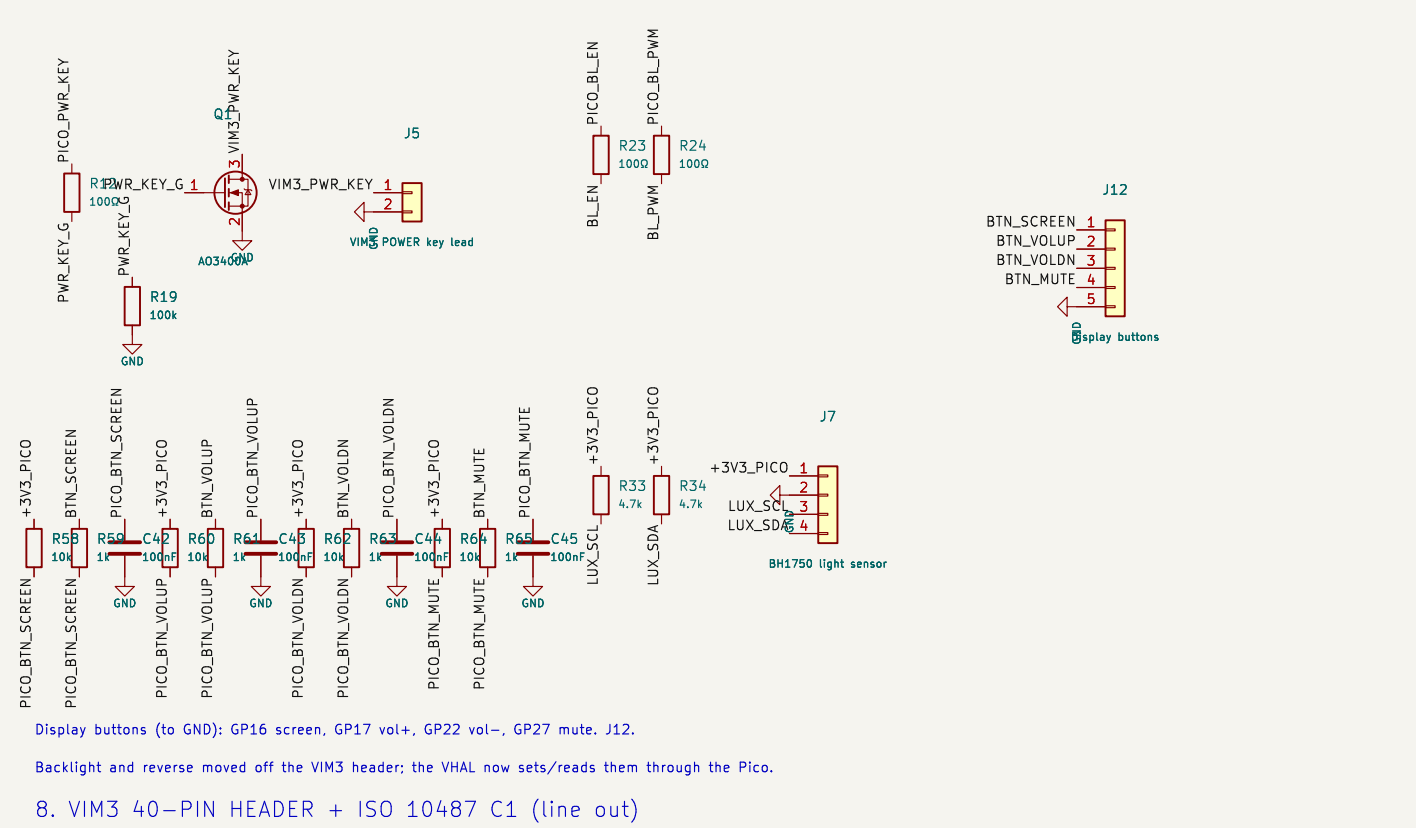
Active antenna (3.3V): VCC_RF -> IOR -> 270 ohm I/O pin. RF_IN is VCC-blocked inside the module (verify in the integration manual).

3. AMPLIFIER REMOTE (+12V from ACC, Pico GP9)

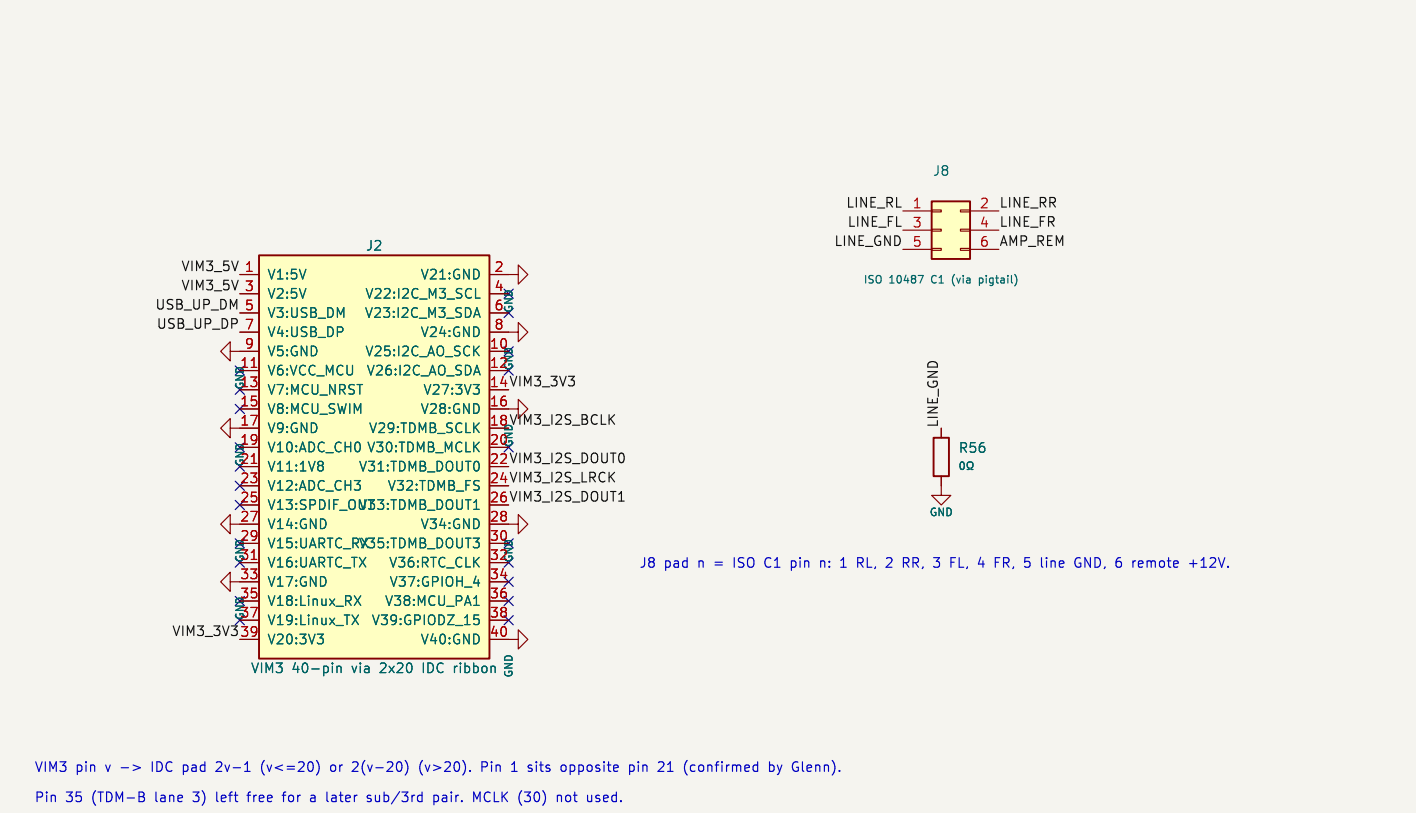


Goes to ISO A3 and C1-6 (A5 via R57). -1A capable; remote inputs draw <100mA.
Only works while ACC is on, even if the Pico hangs.

6. VIM3 POWER KEY, BACKLIGHT, LIGHT SENSOR (all from the Pico)

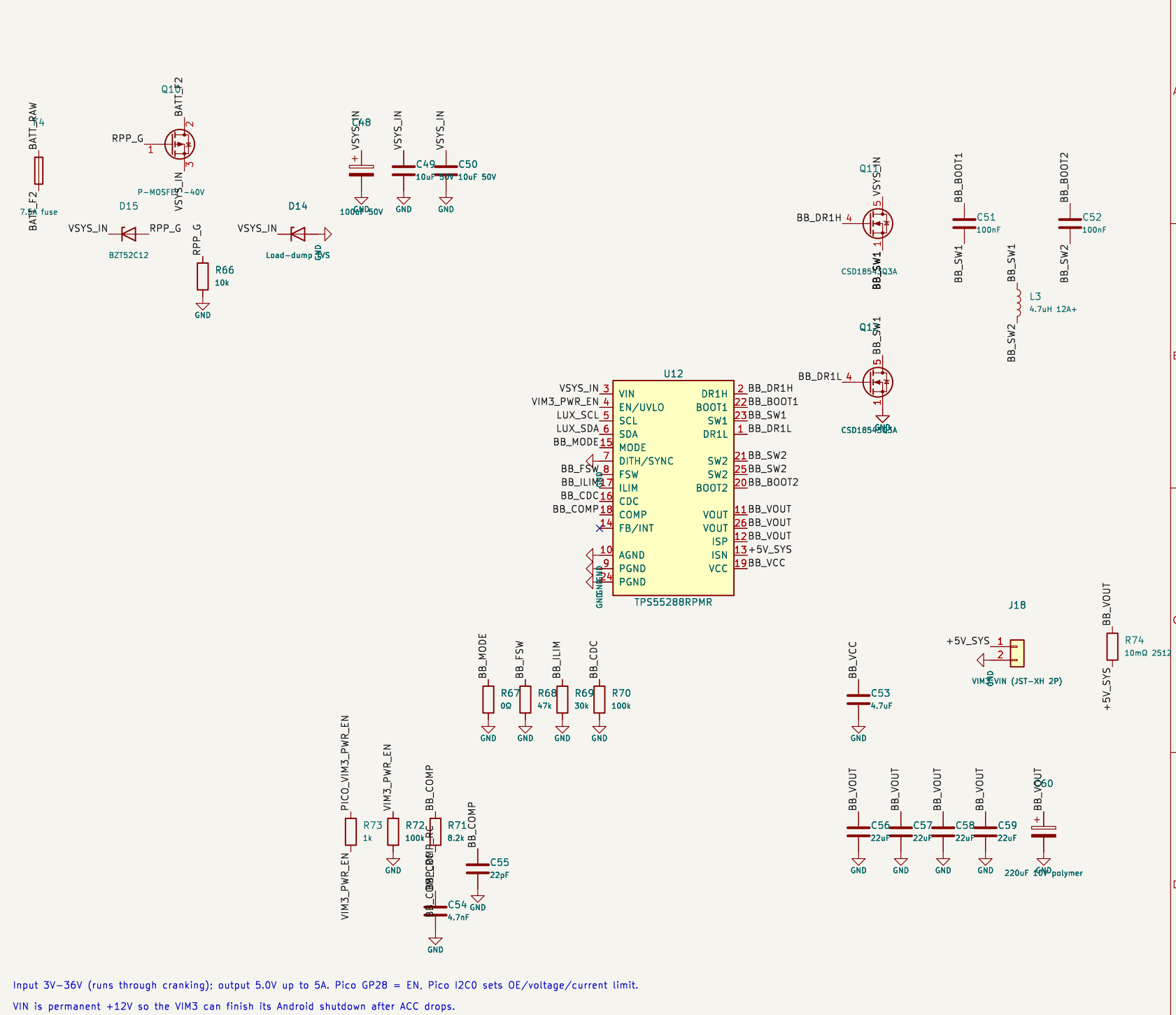


Display buttons (to GND): GP16 screen, GP17 vol+, GP22 vol-, GP27 mute. J12.
Backlight and reverse moved off the VIM3 header: the VIM4L now sets/reads them through the Pico.
8. VIM3 40-PIN HEADER + ISO 10487 C1 (line out)



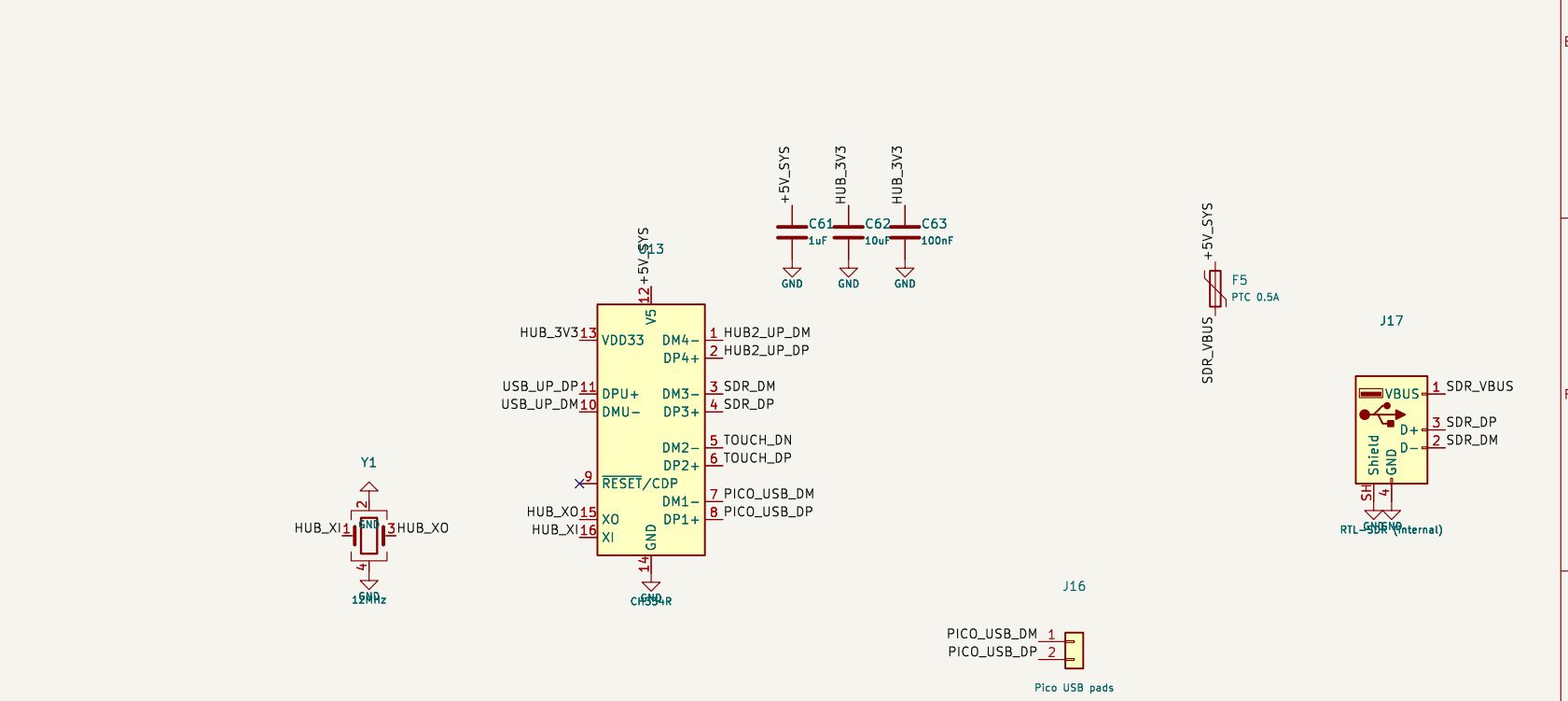
VIM3 pin v -> IDC pad 2v+1 (v+20) or 2(v-20) (v+20). Pin 1 sits opposite pin 21 (confirmed by Glenn).
Pin 35 (TDM-B lane 3) left free for a later sub/3rd pair. MCLK (30) not used.

10. 5V/5A SYSTEM SUPPLY FOR VIM3 + SCREEN + HUB (TP55528B buck-boost, rides through cranking and overvoltage)



Input 3V-36V (runs through cranking); output 5.0V up to 5A. Pico GP28 = EN. Pico i2C0 sets O/V/voltage/current limit.
VIN is permanent +12V so the VIM3 can finish its Android shutdown after ACC drops.

11. USB HUB: VIM3 header USB (pins 3/4) -> Pico, screen touch, spare port



One link to the VIM3: no USB cables. Port 1 Pico (HID + GNSS CDC), 2 touch, 3 R1L-SDR, 4 second hub (external ports). Route 0+/- as 90 ohm pairs.
VIM3 header pins 3/4 are a working USB host port (confirmed by Glenn, 2026-09-29).

Ports are powered only while the VIM3 supply is on. All USB shares the VIM3 header's USB 2.0 link (480 Mbit/s); R1L-SDR needs ~40 Mbit/s.